PULSAR TRIPLE SYSTEM

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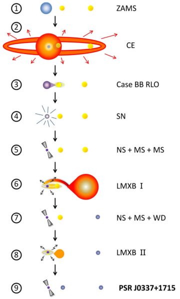
Abstract

Here, we present a self-consistent, semi-analytical solution to the formation of PSR J0337+1715. Our model constrains the peculiar velocity of the system to be less than 160 km/s and brings novel insight to, for example, common envelope evolution in a triple system, for which we find evidence for in-spiral of both outer stars.



Abstract

Table 1
Triple System Parameters at the Onset of Each Stage in Our Scenario (Figure 1) for the Formation of PSR J0337+1715

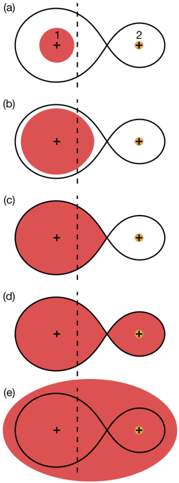
Parameter			Stage								
			1	2	3	4	5	6	7	8	9
Age of the system	t	(Myr)	0.0	23.3	23.3	25.1	25.2	5500	5517	8500	10,500
Mass of primary star	M_1	$(M_\odot)$	10.0	9.90	2.90	1.70	1.28	1.28	1.30	1.30	1.438
Mass of secondary star	M_2	$(M_\odot)$	1.10	1.10	1.10	1.10	1.10	1.10	1.12	1.12	0.197
Mass of tertiary star	M_3	$(M_\odot)$	1.30	1.30	1.30	1.30	1.30	1.30	0.410	0.410	0.410
Orbital period of inner binary	$P_{\mathrm{orb},12}$	(days)	835	849	2.47	0.95	1.55	1.55	1.50	0.90	1.63
Orbital period of tertiary star	$P_{\mathrm{orb},3}$	(days)	4020	4080	17.1	15.7	15.3	14.2	250	250	327
Eccentricity of inner binary	e_{12}		0.00	0.00	0.02	0.01	0.24	0.20	0.02	0.00	0.00
Eccentricity of outer orbit	e_3		0.00	0.00	0.04	0.04	0.22	0.03	0.03	0.03	0.03
Stability parameter	$(R_{\mathrm{peri}}/a_{\mathrm{in}})$		2.96	2.96	3.83	7.07	4.15	4.89	30.8	43.3	35.6
Critical stability limit	$(R_{\mathrm{peri}}/a_{\mathrm{in}})_{\mathrm{crit}}$		2.93	2.93	3.21	3.44	3.93	3.80	3.04	3.04	3.13
Temperature of outer WD	$T_{\mathrm{eff},3}$	(K)							18,000	5800	4300

LMXB and Case BB RLO

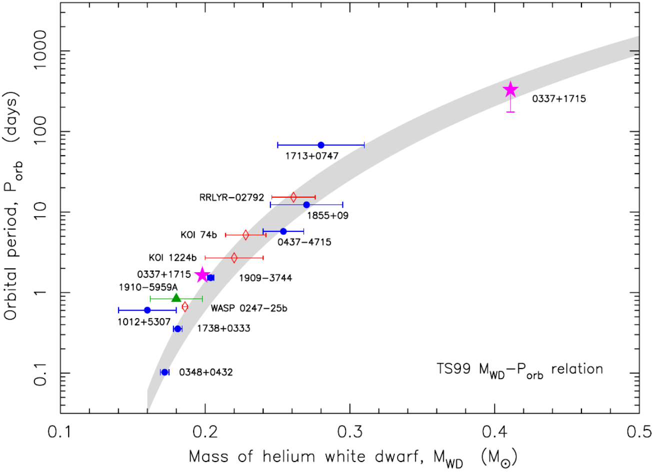
- A low-mass X-ray binary (LMXB) is a binary star system where one of the components is either a black hole or neutron star. The other component, a donor, usually fills its Roche lobe and therefore transfers mass to the compact star. In LMXB systems the donor is less massive than the compact object.
- A Case BB RLO is a type of Roche-lobe overflow (RLO) initiated by the helium star expansion after core helium exhaustion.

Common Envelope (CE)

A common envelope is formed in a binary star system when the orbital separation decreases rapidly or one of the stars expands rapidly. The donor star will start mass transfer when it overfills its Roche lobe and as a consequence the orbit will shrink further causing it to overflow the Roche lobe even more, which accelerates the mass transfer. This phase of the shrinking of the orbit inside the common envelope is known as a spiral-in.



Support to the M_{WD}–P_{orb} Relation



Eccentricities larger than expected

- How to observe the eccentricities:

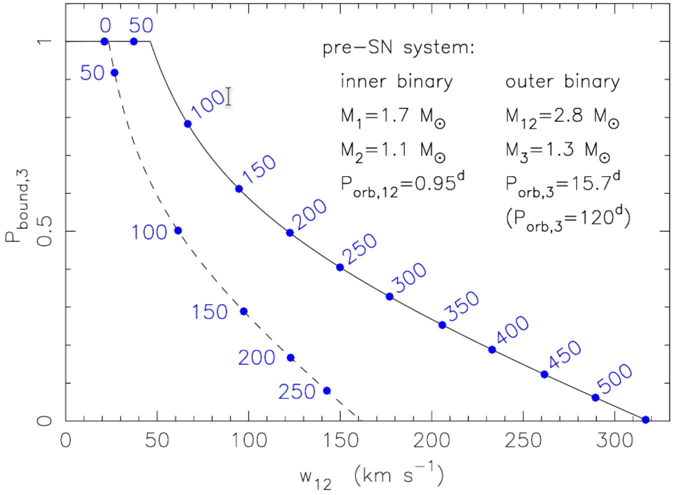
$$\delta t = \Delta t - \Delta t_{\text{circ}} = \frac{ea \sin i}{2c} \left[\sin \omega + \sin\left(\frac{2\pi t}{P} - \omega \right) \right]$$

? What determines the eccentricities and why are the eccentricities larger?

Evolution of the Two LMXB Phases

- Both LMXB phases evolved highly nonconservatively.
- The changes of the orbital separations: in the most extreme case (a pure fast, or Jeans mode wind mass loss from the inner binary) $P_{\mathrm{orb},3}$ could be as short as 175 days.
- Major uncertainties:
  - spin–orbit couplings mediated by tidal torques.
  - accretion onto the inner binary system during mass transfer from the tertiary star

The Dynamical Effects of the SN Explosion



Common Envelope Evolution

- CE evolution (stage 2) not only leads to efficient orbital angular momentum loss of the inner binary orbit, also the tertiary star is subject to efficient in-spiral.
  - the post-SN $P_{\mathrm{orb},3}$ (after recircularization) must match the expected orbital period at the onset of the outer LMXB phase.
  - to avoid a very small survival probability as a consequence of the SN.
- The lower limit of $P_{\mathrm{orb},3}$ is set at about 4000 days on the ZAMS.

Alternative Models

- The outer WD may form last.
- It was formed in a quadruple system where the SN kick caused interactions with an outer binary and ejection of the fourth member.
- It evolved in a globular cluster and subsequently ejected into the Galactic field.

Discussion

- ? How to calculate the critical stability limit?
- ? How is a ”inner binary kick”?

Thank you for listening!